

## SE Lab Assignment 2 (30% of QT2(Process Mark 2)): 4 weeks (from Lab 7- Lab10)

### Winform App/Web-App/Github: 2 students work together.

Given the following data tables:

- Item(ItemID, ItemName, Size,.....)
- Agent(AgentID, AgentName, Address)
- Order(OrderID, OrderDate, AgentID)
- OrderDetail(ID, OrderID, ItemID, Quantity, UnitAmount)
- Users(UserID, UserName, email, password, lock,...)

*(Insert 15-30 rows data for each table)*

#### **Scenario:**

- Allow users to enter data about items, agents
- Allow user to enter order when agent want to order items, one agent can order multiple items in an order.

#### **Exer1: Windows and Web Form Application Project (3-Tiers Architecture)**

1. Login Form to Main Form
2. Create Windows Form for adding products, agents and a order form (one-many) (Order and OrderDetail)
3. Create a report for printing the order
4. Create a form for filter best items, items purchased by customers, customer purchases items

**Exer2: ASP.Net MVC Project (.Net Framework) (DB First)** (remove all irrelevant items from template project)

1. Login Form to Main Form
2. Create web form for adding products, agents and a order form (one-many) (Order and OrderDetail)
3. Create a button on Order form then link to the order form (to print) or report
4. Create a web form for filter best items, items purchased by customers, customer purchases items

#### **Exer3: ASP.NET Web App Core with Razor Project** (either DB First or Code First)

- Same requirements with Exer2

**Exer4: ASP.NET Web App Core (MVC) (Code First)**, Define models and data migration then create Controller with View, use Entity Framework. **Same requirements with Exer2**

- <https://learn.microsoft.com/en-us/aspnet/core/data/ef-mvc/intro?view=aspnetcore-9.0>
- <https://www.c-sharpcorner.com/article/code-first-approach-in-asp-net-core-mvc-with-ef-core-migration/>

**Exer5:** Create a Unit Test Project to test those 3 projects (Exer1,2,3).

**Notes:**

- Use App.Config for Connection String (refer to given previous Project)
- Also recommend to Create MVC Web, bootstrap for icon, format,...
- Create a repository for those either 3 or 4 projects on Github

**Submit:**

- Report.docx (Make a proper report with screenshot and explain for each exercise) . Last page is a contribution of each team member
- .zip (source code) or Github link
- Readme.txt (contain link : github, video demo (if any), how to run/combine your source code)

You should complete at least one of Exer2,3,4 to gain your knowledge in order to conduct your Final Project.

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